



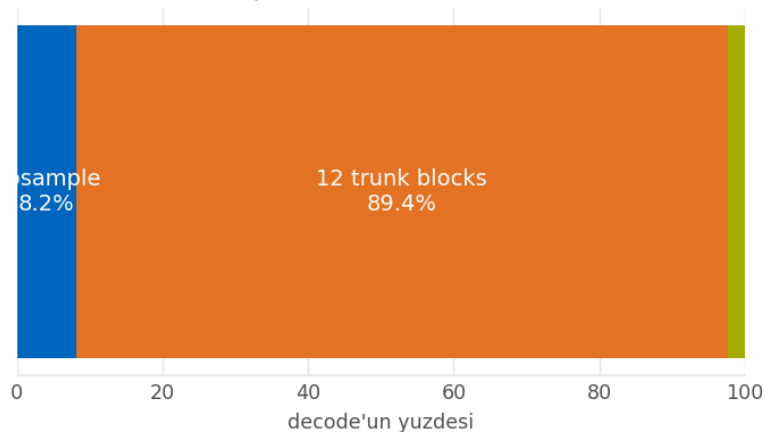
FLEX-UF: Content-Adaptive Early Exit for DCVC-UF

Decoder compute saved at a measured cost in quality
Chair of Media Technology · TUM

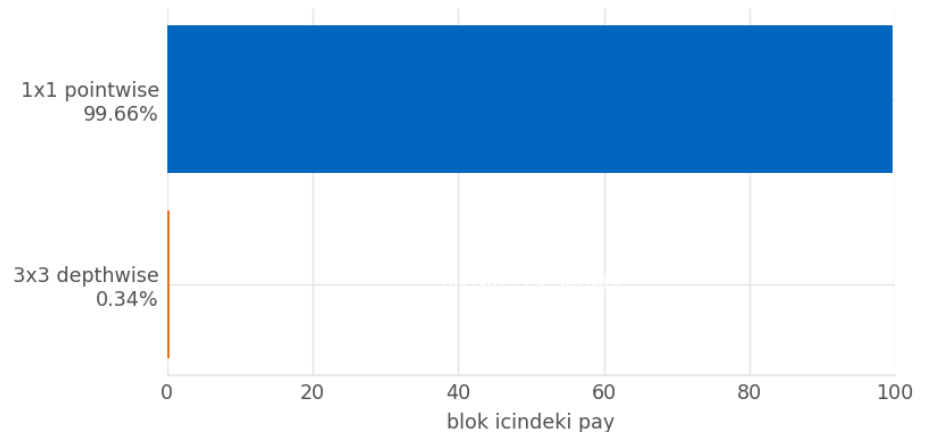
Problem and goal

- **DCVC-UF's intra decoder: 453.5 GMAC per 1080p frame**
 - ☐ 12 identical DepthConvBlocks = 89.4% of it
 - ☐ every patch pays the same, however easy it is
- **Goal: spend less compute where the content allows**
 - ☐ target set with the group: 30–40% saved at ≤ 0.1 dB
- **Constraint that shapes everything: the reference is the RELEASED model**
 - ☐ encoder, hyperprior, entropy model frozen $\rightarrow$ identical bitstream
 - ☐ so any measured difference is the decoder alone

453.5 GMAC / 1080p kare



Bir DepthConvBlock icinde

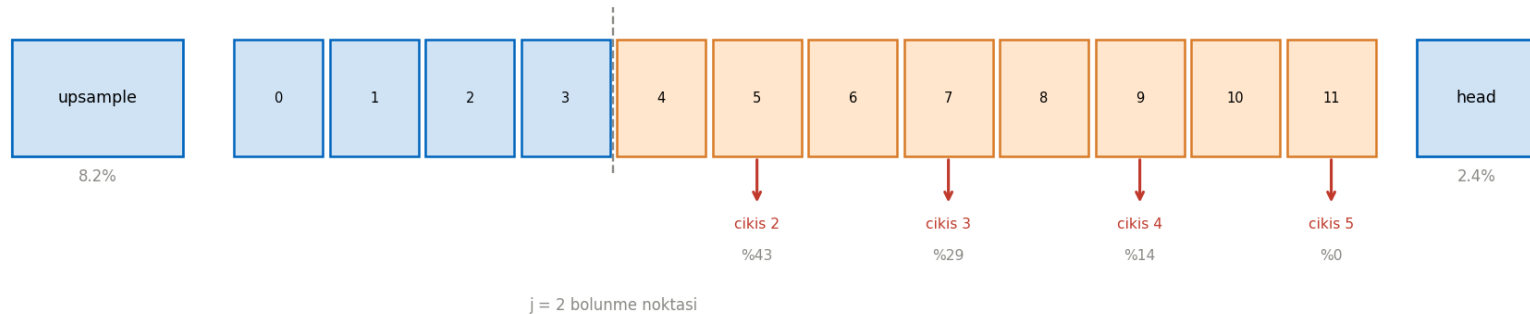


Architecture: a ladder of exits

- **Shared stem runs full-frame → no seams; the rest runs per tile**
- Each exit has a zero-initialised 1×1 adapter, then the shared head
- Pointwise on purpose: a 1×1 adds no receptive field, so no seam cost

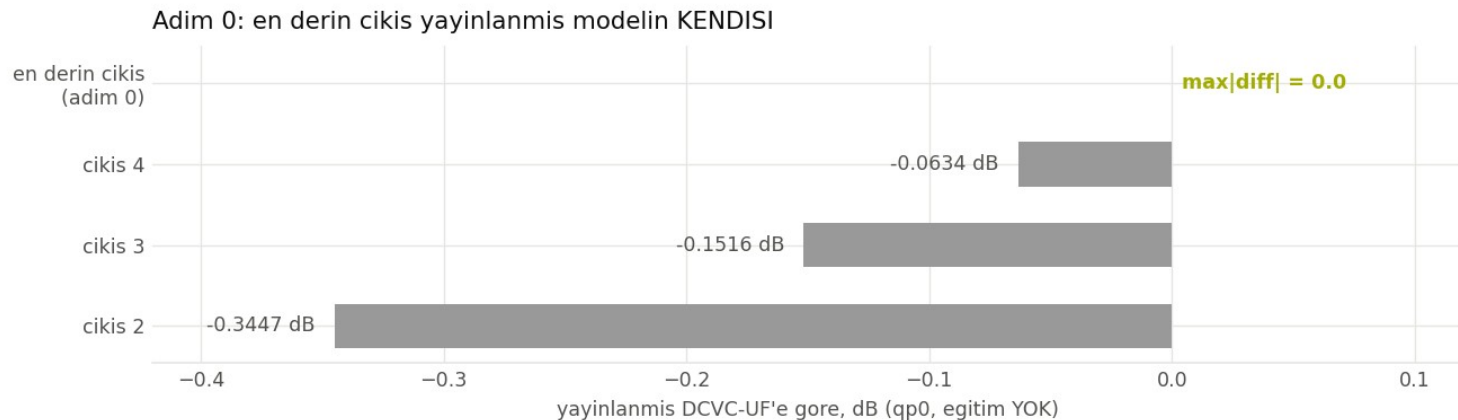
tam kare (dikis yok)

tile basına — atlanabilen kısım, decode'un %89.4'u



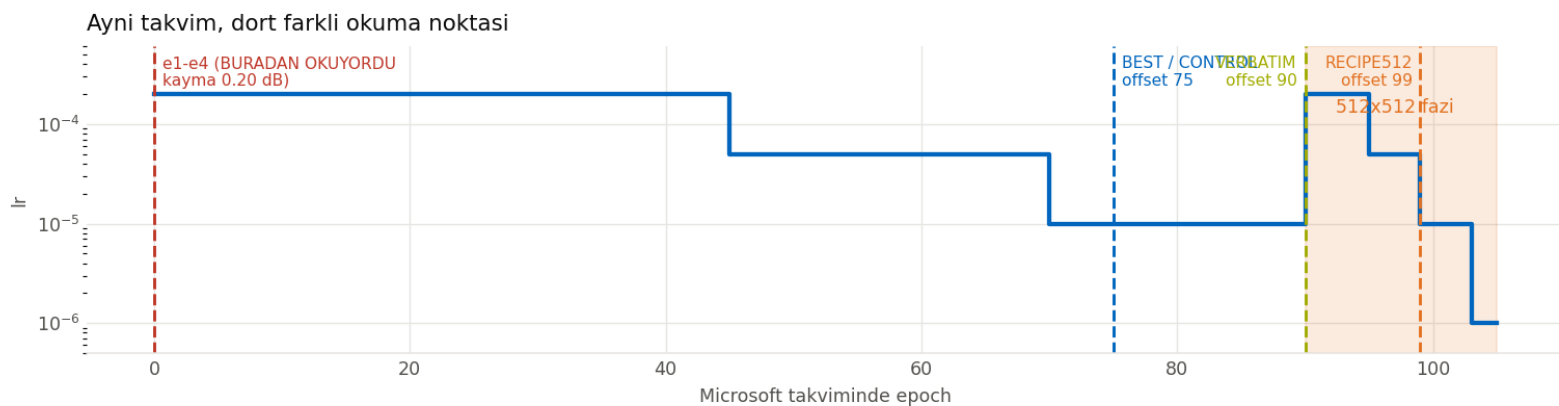
We start AT DCVC-UF, not near it

- **The released weights are re-expressed in ladder form by a key remap**
 - `dec_1.0` → upsample, `dec_1.n` → groups.g.i, `dec_2` → head
 - the remap is a bijection over 143 tensors (asserted in tests)
- Adapters and seam repair are zero-init, i.e. the identity
- **So at step 0, with no training at all:**
 - released DCVC-UF vs our deepest exit → $\max|\text{diff}| = 0.0$
- **Training does not have to reach DCVC-UF. It has to (a) lift the shallow exits and (b) not damage the deepest one.**
- From-scratch runs were abandoned for exactly this: after 6 epochs their anchor sat 1.49 dB below the release and was not closing.



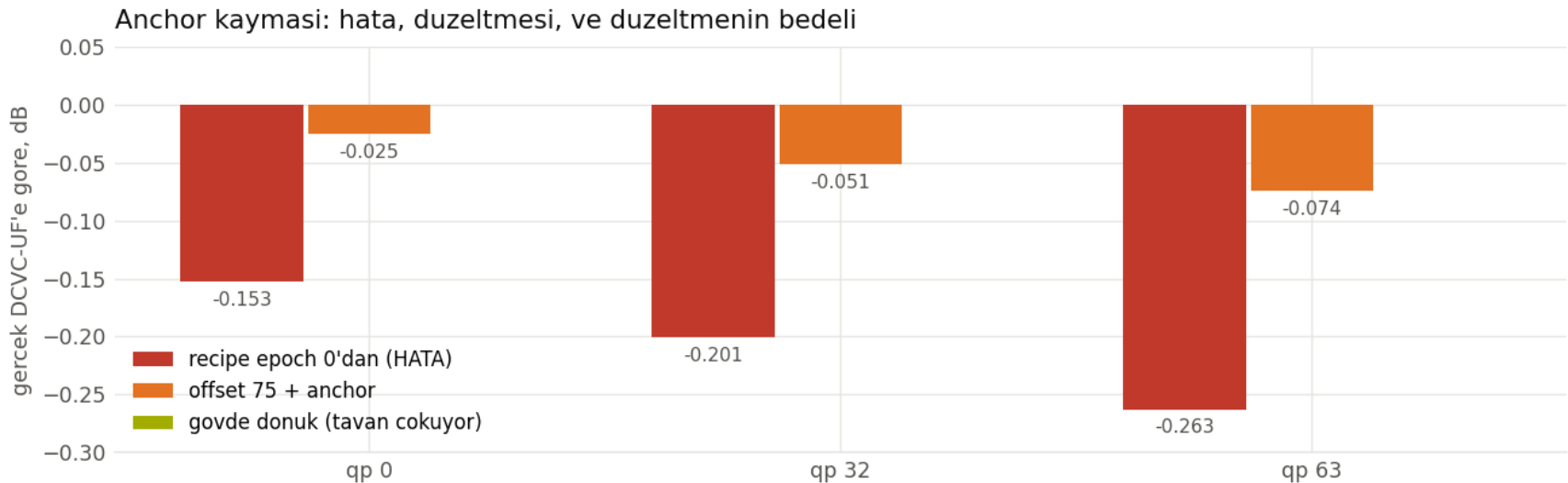
Training: the recipe, checked not claimed

- **scripts/verify_recipe.py compares each item against ~/DCVC/train_image.py and exits non-zero on a mismatch**
 - ☐ 8 schedule rows character for character · 106 entries
 - ☐ AdamW 1e-4 · clip_grad_norm 0.1 · non-finite batch skipped
 - ☐ ImageFolder + get_training_lambdas · 64 QP levels
 - ☐ encoder identical over 74 tensors · 255 shared tensors, no shape change
- **Deliberate additions, listed rather than hidden:**
 - ☐ --epoch_offset (read the schedule where a warm start actually is)
 - ☐ --anchor_weight, --new_lr_scale, --freeze_encoder
- **VERBATIM run: their final phase (epochs 90–104, 512px, lr 2e-4 → 1e-6) with NONE of those additions, effective batch 16 by accumulation**



The anchor, and a $\times 4$ error caught

- **Reading the schedule from epoch 0 applied the from-scratch lr to a converged model**
 - deepest exit fell 0.153 / 0.201 / 0.263 dB in ONE epoch (qp0/32/63)
 - fixed to 0.025 / 0.051 / 0.074 by --epoch_offset 75 + --anchor_weight
- **Freezing the backbone gives exactly 0.000 — but collapses the ceiling from 27.3% to 11.5%, so training the trunk is mandatory**



What an exit costs, on one frame

- **Bosphorus 1080p, qp32, identical bitstream at every exit**
- Error maps $\times 25$: the damage is structured, concentrated on detail and on tile borders — which is what the seam work targets

Bosphorus 1080p, qp32 — aynı bitstream, farklı çıkışlar

yayınlanmış DCVC-UF



cikis 2
%43 tasarruf, +1.083 dB



cikis 3
%29 tasarruf, +0.582 dB



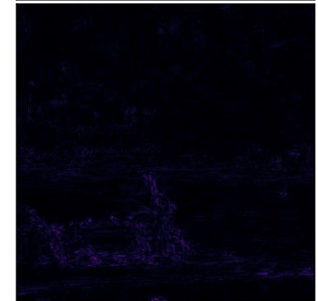
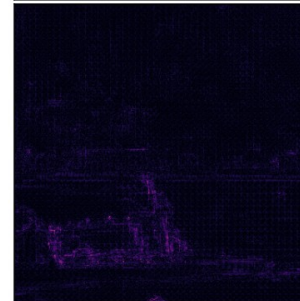
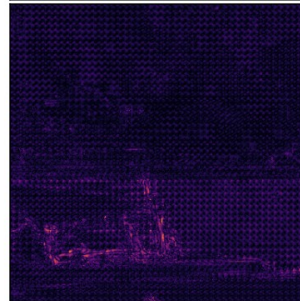
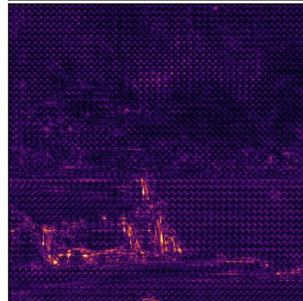
cikis 4
%14 tasarruf, +0.125 dB



cikis 5
%0 tasarruf, +0.033 dB

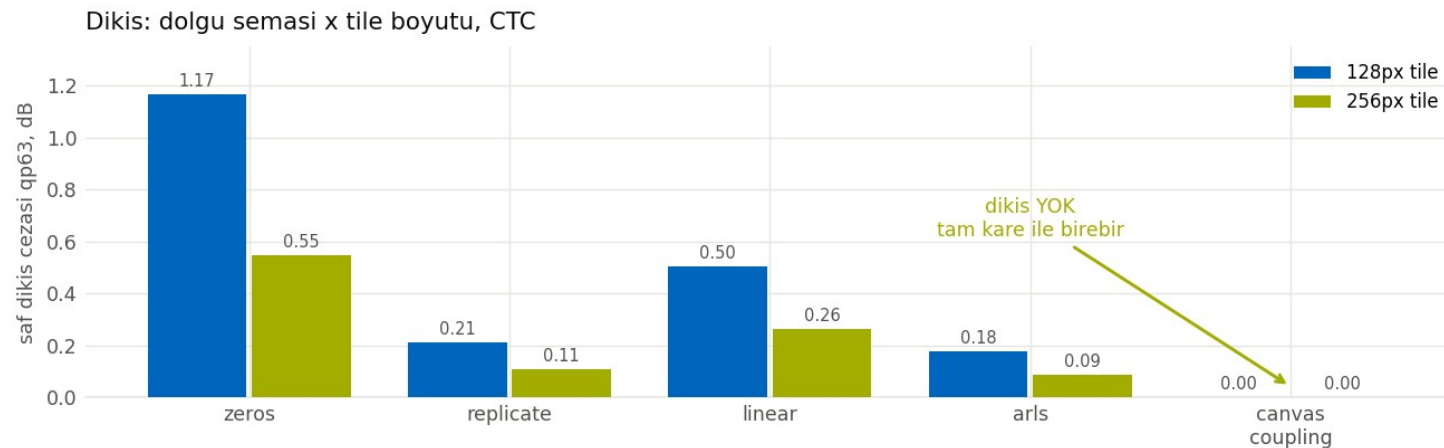


hata haritaları
($\times 25$ güçlendirilmiş)



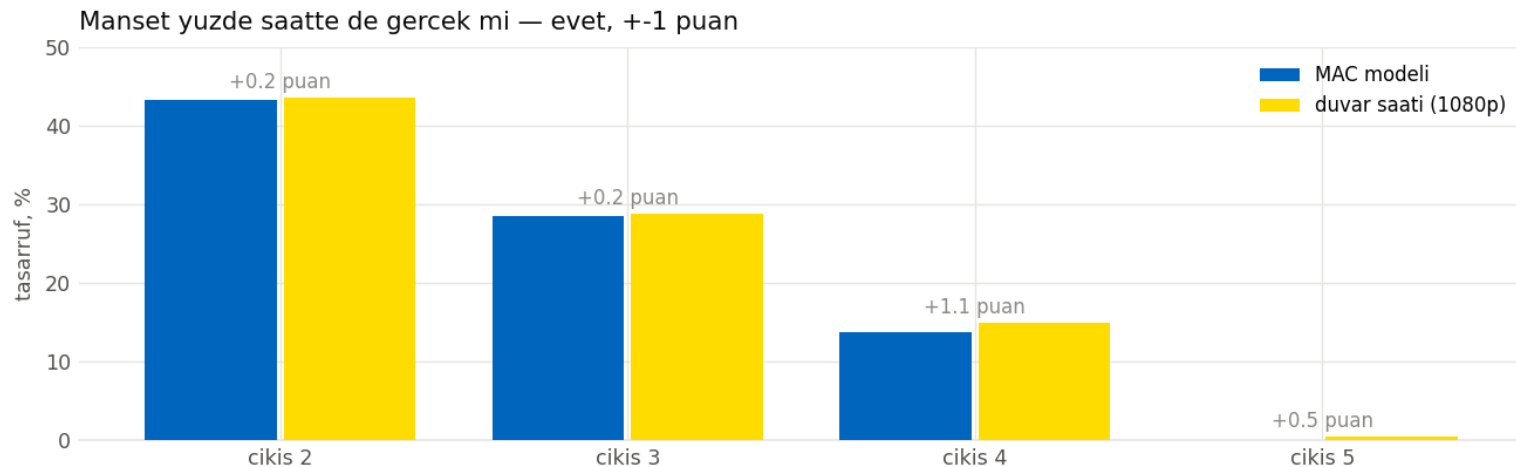
The seam, and how it was removed

- **A tile decoded alone meets its border with invented values**
- **Pure seam penalty at qp63, CTC, no early exit taken:**
 - zeros 1.167 dB · replicate 0.213 · arls 0.179 · 256px halves each
- **arls (arXiv:2502.12300) was implemented, measured, and REJECTED**
 - +0.034 dB for +10.7% of decode wall-clock — a bad trade
- **Canvas coupling: only the 3×3 depthwise has spatial extent — 0.334% of a block — so only it needs the neighbour**
 - cost +0.03% of the decode; earlier halo work paid 26% by haloing the whole block, which was the wrong thing to halo
 - at uniform depth: full-frame vs tiled → $\max|\text{diff}| = 0.0$
- **The seam does not shrink. It stops existing.**



Cost, checked in the right unit

- **The headline is a percentage; it comes from a MAC model**
- **MACs are right for a paper and wrong for a promise, so wall-clock was measured on a 1080p decode:**
 - exit 2: 43.4% (MAC) vs 43.6% (clock)
 - exit 3: 28.6% vs 28.8% · exit 4: 13.8% vs 14.9%
- **Agreement within ± 1 point at every exit**
- The same question caught arls: benefit in dB, bill in milliseconds
- And the cost model itself is checked against the executed decode (tests/test_cost_matches_reality.py)

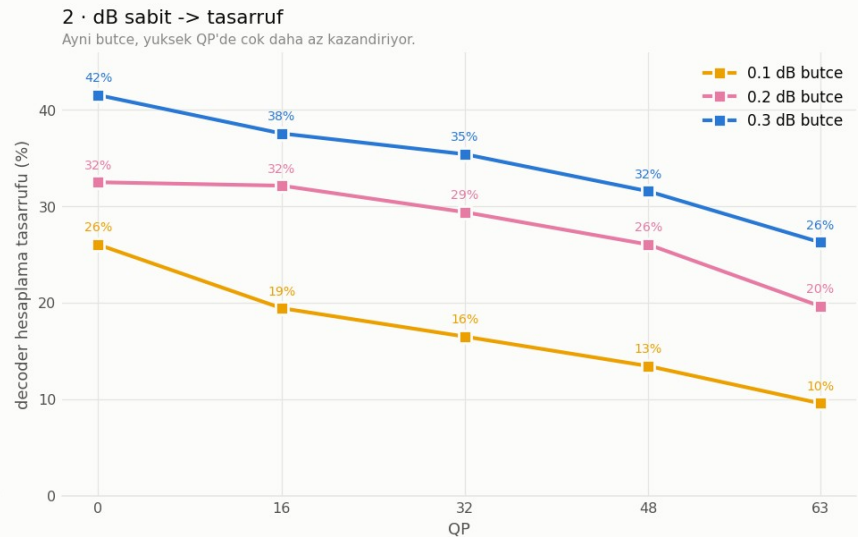
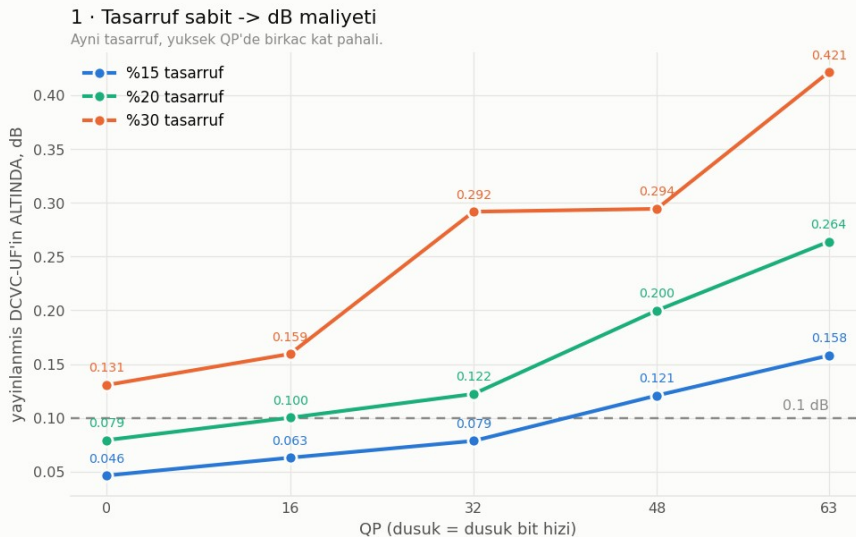


Results vs the released decoder

- **At 0.3 dB: 42% (qp0) · 35% (qp32) · 26% (qp63)**
- At 0.1 dB: 26% · 16% · 10% — the 30–40% target is met at 0.3 dB, not at 0.1
- Saving falls with rate because early exit costs more there: the exit-2 penalty grows 3.4× from qp0 to qp63 as the latent carries more detail

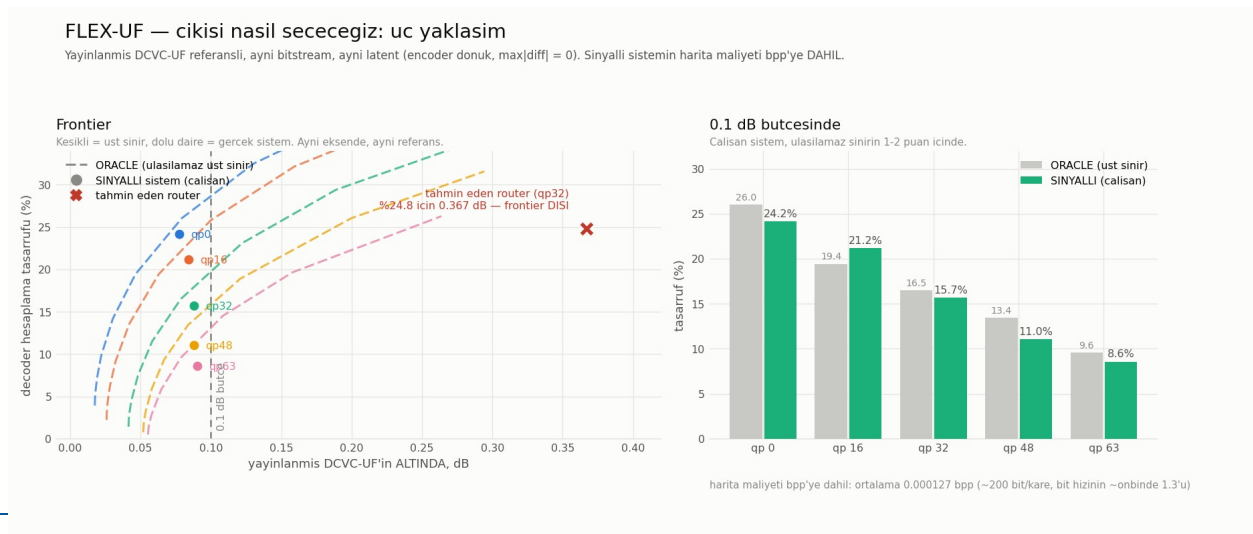
FLEX-UF — aynı yüzey, iki kesit

Yayınlanmış DCVC-UF referanslı, CTC, aynı bitstream ve latent (encoder donuk, $\max[\text{diff}] = 0$). ORACLE: kusursuz seçim varsayımı.



Router: predicting failed, signalling works

- **A learned router could not reach the oracle, and 12k steps showed why**
 - CE $0.5 \rightarrow 0.09$ while qp0 agreement moved $0.743 \rightarrow 0.741$, regret unchanged
 - fitting the training data $4\times$ better and behaving identically on test is missing INFORMATION, not missing capacity
- **Structural: the oracle uses error against the SOURCE; the decoder never sees the source**
- **But the encoder does — and in video coding mode decisions are signalled, not inferred (HEVC/VVC send partitioning and prediction mode)**
 - exit map costs $1.3e-4$ bpp entropy-coded — 4 orders below the frame
 - measured with the map's cost INSIDE the bitrate: 24.2 / 21.2 / 15.7 / 11.0 / 8.6% at qp0...63, all under 0.1 dB
- **Agreement with the oracle becomes 100% by construction**



Theory, and where we are

- **Cost is additive over tiles and MSE is a mean over tiles → both affine in the assignment**
 - fixed proportions reach exactly the convex hull of the K exit points
 - per-tile assignment reaches the Minkowski average of the N tile hulls
 - their gap is min-of-average – average-of-min ≥ 0 , zero iff every tile prefers the same exit — this IS the value of adaptivity
- **Measured: Δ/J rises 0.68% → 4.52% with rate**
- **Open, and stated as open:**
 - 0.1 dB target not met; 0.3 dB comfortably met
 - larger tiles cut the seam but also the adaptivity — being isolated now
 - HEVC classes B/C/D behind JVET credentials, reported as NOT MEASURED
- 4 runs training (BEST, CONTROL, RECIPE512, VERBATIM) on 4 GPUs

